

## **Final Project: Proposal**

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The problem I want to address is one I've dealt with for many years: Weight loss. Weight loss itself isn't the problem; it's sticking to a routine and actually losing weight. The CDC reports that 49.1% of adults try to lose weight each year. Research also shows that most people fail to lose weight: they lose only about 10% of their body weight, then regain it within 1 year and all their weight within 5 years, according to Health Psychology Open. A lot of times, we know why we aren't losing weight: “too many calories”, “not enough protein”, “too many carbs”, and not enough exercise, and while these could all be true, we are missing another reason, which is lack of planning and organization. An article by RELATED mentions “A recent study has found that 'stressful and chaotic food environments' influence people to reach for high-calorie snacks.” While this study relates to a physically chaotic environment, it suggests that this sense of chaos triggers a brain response that impels you to develop “bad” eating habits. The application would be a platform where a user could get an organized meal plan, exercise routine, and timeline. The target users are people like me who struggle to lose weight due to a lack of structure and planning, or about 50% of Americans. An AI solution addresses this problem by being cost-effective, readily available, and productive. In contrast, in real life, you would need a lot of money for a medical professional, a trainer, and a dietitian.

For beginners, a user would enter their goals in plain language, such as “I want to lose 20 pounds and gain muscle.” The agent will then ask for their weight, height, exercise preferences, and meal preferences. The agent will then create a specialized weekly meal plan and exercise routine. Let's also say the user lacks vision and wants to know how fast the process will take if they follow the routine strictly. Based on the user's earlier inputs, the

agent can estimate how long it will take to notice significant progress. There could also be a plateau in weight loss, and here is where the integrated AI chatbot comes in handy: the agent can present web solutions to the user and help them select one to apply to and edit their weight-loss plan.

For the initial system design, I would need many agents. A planner agent that will create a structured weight-loss plan based on users' goals and personal preferences. This planner would determine nutritional intake and calorie targets, and generate a weekly structure. A calculator agent that performs numerical calculations, such as BMI, macro intake, estimated weight-loss timeline, and plateau detection. An external data agent that pulls information from reliable sources to address users' weight-loss problems. A summarizer agent that takes health data and converts it into simple plans users can use immediately. Lastly, an Evaluating agent that evaluates a user's progress, detects plateaus, and tweaks plans by suggesting improvements. I expect to implement many tools, such as Google Docs for prompt writing, Grammarly for grammar checks, Opal for app development, health statistics, user input, nutrition databases, OpenAI for touch-ups in certain areas, and CDC health information.

Potential challenges include hallucinations within the app; they might generate incorrect nutrition, workout, or health advice. To combat this, I would limit responses to structured prompts, use verified data from credible sources, and include a disclaimer. API failures can also occur if the API doesn't return the correct information to the agent. To avoid this, I can include fallback responses and a retry message. Constraints for my project include rate limits, as I may run into request limits. Development constraints are intimidating because there is limited time, and I have limited knowledge of development. The most significant constraint is privacy concerns about individuals' weight and health information. To battle these concerns I plan to perfect my prompts efficiency to decrease the amount of request, for

development I plan to do basic progress adjustments and work little by little as well as learn from my peers and professor, and lastly regarding the privacy concerns I would suggest the application store the information anonymously and leave a disclaimer for the website that for any significant health concerns may need to be checked with a licensed professional.

Lastly, my timeline is 4-5 weeks, with Week 1 consisting of planning and researching everything I will need to complete my project and gathering multiple sources to link to my agent and create prompts. Week 2 will be developing the MVP and tweaking the agents required to connect the user's input to the workout plan generation. Week 3 will consist of pulling the final pieces together to ensure my MVP is where I want it to be. Week 4 will finally test the outputs of the planning and calculating agents, fixing and correcting any errors I may have, as well as adding plateau detection. Week five is the final week, when I will finalize the opal app, prepare the demo, and ensure there are no errors so it's up and working.

### **Citations**

<https://www.singlecare.com/blog/news/weight-loss-statistics/>

<https://www.prevention.com/weight-loss/a20513150/how-to-get-organized-lose-weight-faster/>